Preprocess — Mon May 25 26, 12:45:54

Settings

Continuize Discrete Variables: One feature per value
Normalize Features: Standardize to $\mu=0$, $\sigma^2=1$

Random Forest — Mon May 25 26, 12:45:59

Name: Random Forest

Model parameters

Number of trees: 100
Maximal number of considered features: unlimited
Replicable training: Yes
Maximal tree depth: unlimited
Stop splitting nodes with maximum instances: 5
Class weights: True

Data

Data instances: 5820
Features: instr=1, instr=2, instr=3, class=1, class=2, class=3, class=4, class=5, class=6, class=7, class=8, class=9, class=10, class=11, class=12, class=13, nb.repeat=1, nb.repeat=2, nb.repeat=3, attendance=0, attendance=1, attendance=2, attendance=3, attendance=4, Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q15, Q16, Q17, Q18, Q19, Q20, Q21, Q22, Q23, Q24, Q25, Q26, Q27, Q28 (total: 52 features)
Target: difficulty

Test and Score — Mon May 25 26, 12:46:23

Settings

Sampling type: 5-fold Cross validation
Target class: None, show average over classes

Scores

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.727	0.468	0.468	0.468	0.468	0.305

Input

Features: instr=1, instr=2, instr=3, class=1, class=2, class=3, class=4, class=5, class=6, class=7, class=8, class=9, class=10, class=11, class=12, class=13, nb.repeat=1, nb.repeat=2, nb.repeat=3, attendance=0, attendance=1, attendance=2, attendance=3, attendance=4, Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q15, Q16, Q17, Q18, Q19, Q20, Q21, Q22, Q23, Q24, Q25, Q26, Q27, Q28 (total: 52 features)
Meta attributes: Random Forest, Random Forest (1), Random Forest (2), Random Forest (3), Random Forest (4), Random Forest (5), Fold
Target: difficulty

Ranks

		#	Gain ratio	Gini
1	N attendance=0		0.330	0.111
2	N class=7		0.083	0.005
3	N attendance=3		0.072	0.017
4	N attendance=1		0.063	0.011
5	N attendance=2		0.058	0.011
6	N attendance=4		0.046	0.008
7	N nb.repeat=2		0.041	0.005
8	N nb.repeat=1		0.038	0.006
9	N class=6		0.037	0.004
10	N class=2		0.034	0.002
11	N class=3		0.033	0.005
12	N class=12		0.032	0.001
13	N instr=2		0.028	0.006
14	N Q17		0.027	0.019
15	N class=1		0.025	0.002
16	N instr=3		0.025	0.008
17	N Q22		0.024	0.017
18	N Q13		0.023	0.015
19	N Q19		0.022	0.015
20	N nb.repeat=3		0.022	0.002
21	N Q25		0.022	0.016
22	N Q3		0.022	0.014
23	N Q15		0.022	0.015
24	N Q14		0.022	0.015
25	N Q28		0.022	0.015
26	N Q21		0.022	0.015
27	N class=13		0.022	0.004
28	N Q26		0.021	0.013
29	N Q20		0.020	0.014
30	N Q11		0.020	0.013
31	N Q23		0.020	0.013
32	N Q18		0.020	0.013
33	N Q9		0.019	0.012
34	N Q12		0.018	0.011
35	N Q5		0.018	0.012
36	N class=10		0.018	0.002
37	N Q2		0.018	0.012

38	N	Q10	0.018	0.011
39	N	Q16	0.017	0.011
40	N	Q6	0.016	0.011
41	N	Q7	0.016	0.011
42	N	Q1	0.016	0.011
43	N	Q24	0.016	0.011
44	N	Q4	0.015	0.010
45	N	Q27	0.015	0.010
46	N	instr=1	0.014	0.003
47	N	class=5	0.014	0.002
48	N	Q8	0.013	0.008
49	N	class=11	0.013	0.002
50	N	class=4	0.008	0.000
51	N	class=8	0.006	0.001
52	N	class=9	0.005	0.000

Confusion Matrix

Mon May 25 26, 12:46:34

Confusion matrix for Random Forest (showing number of instances)

		Predicted					
		1	2	3	4	5	∑
Actual	1	1243	63	134	77	103	1.620
	2	71	115	238	77	48	549
	3	153	231	760	436	194	1.774
	4	95	104	414	424	188	1.225
	5	89	57	169	158	179	652
	∑	1.651	570	1.715	1.172	712	5.820